

Mathematical Physics and Computation

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September 9, 2026

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Linear Operators

Change of basis Let V be a finite-dimensional vector space of dimension n . Let $\beta = \{e_1, \dots, e_n\}$ and $\beta' = \{e'_1, \dots, e'_n\}$ be both basis for V . We are interested in change representation of $x \in V$ from using β to β' . To do this we need to know how $e_j \in \beta$ can be represented using β' :

$$e_\nu = a_\nu^\mu e'_\mu \quad (1)$$

Here coefficients a_ν^μ can be organized into a matrix. The matrix a_ν^μ looks like:

$$\begin{pmatrix} a_1^1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & & \vdots \\ a_n^1 & a_n^2 & \cdots & a_n^n \end{pmatrix} \quad (2)$$

Contravariant Transformation There are two interpretations of a_ν^μ :

1. (a_ν^μ transform β' to β) This is the definition above:

$$e_\nu = a_\nu^\mu e'_\mu \quad (3)$$

2. (a_ν^μ transform representation in β to β') Now for arbitrary $x \in V$, we have:

$$x = c'_\nu e'_\nu = c_\nu e_\nu = c_\nu (a_\nu^\mu e'_\mu) = (c_\nu a_\nu^\mu) e'_\mu \quad (4)$$

from which we recognize:

$$c'_\nu = (c_\nu a_\nu^\mu) \quad (5)$$

We note that the matrix has *opposite* effect on basis and representation. This is actually intuitive: in order to represent the vector in new basis β' , we will have to write the old basis $e_\nu \in \beta$ using linear combinations of the new basis e'_μ . The later which effectively transforms basis in β' to β . For this reason, the representation is said to be transformed **contravariantly**.

Notation

The matrix a_ν^μ is commonly denoted by $[I]_\beta^{\beta'}$ in math textbooks. Therefore:

- Transform in representation is denoted by:

$$[x]_{\beta'} = [I]_\beta^{\beta'} [x]_\beta \quad (6)$$

- Transform in matrix representation of operator is given by:

$$[T]_\beta = [I]_{\beta'}^\beta [T]_{\beta'} [I]_\beta^{\beta'} \quad (7)$$

Triangularization

Proposition 0.1.1 (Existence of eigenvalue in vector spaces on algebraically closed field). *Suppose V is a finite-dimensional vector space over an algebraically closed field $\mathbb{F}$, and $T \in \mathcal{L}(V)$, then T has at least one eigenvalue*

Theorem 0.1.1 (Algebraically closed field guarantees triangularizability). *Suppose V is a finite-dimensional vector space over an algebraically closed field $\mathbb{F}$, and $T \in \mathcal{L}(V)$, then T is always triangularizable.*

Inner Product Space

Note: The development of concepts need to keep extension to infinite analog in mind

The identity operator and the Kroncker's Delta Let V be a vector space endorsed with an inner product $\langle \cdot | \cdot \rangle$ of dimension n and $\beta = \{e_\nu\}$ be an orthonormal basis. Then we can compute the inner product between arbitrary $x, y \in V$ as follows:

$$\begin{aligned}\langle x|y\rangle &= \langle e_\nu|x\rangle^* \langle e_\nu|y\rangle \\ &= \langle x|e_\nu\rangle \langle e_\nu|y\rangle \\ &= \langle x||e_\nu\rangle \langle e_\nu||y\rangle\end{aligned}\tag{8}$$

But $\langle x|y\rangle = \langle x|I|y\rangle$, hence we get that

Lemma 0.1.2 (Outer-product decomposition of the identity operator). *Let $\{e_\nu\}$ be an orthonormal basis for V , then:*

$$|e_\nu\rangle\langle e_\nu| = I\tag{9}$$

Help! What is happening here?

Step 1 We first rewrite inner product as a sum of component-component product. In $\mathbb{C}^3$ for instance,

$$\begin{aligned}\langle x|y\rangle &= \sum_\nu (x^\nu)^* y^\nu \\ &= \sum_\nu \langle e_\nu|x\rangle^* \langle e_\nu|y\rangle\end{aligned}\tag{10}$$

Note in the last step we envicted orthonormality of $\{e_\nu\}$. Adopting Einstein notaiton we get the first equality.

Step 2 We are bascially writing two consecutive inner product $\langle x, \alpha \rangle \langle \beta, y \rangle$ as a bilinear form $\langle x, Ay \rangle$ where $A = |\alpha\rangle\langle\beta|$. Consider this example in $\mathbb{R}^{\mathbb{K}}$:

$$\langle x, \alpha \rangle \langle \beta, y \rangle = \begin{pmatrix} x^1 & x^2 & x^3 \end{pmatrix} \begin{pmatrix} \alpha^1 \\ \alpha^2 \\ \alpha^3 \end{pmatrix} \begin{pmatrix} \beta^1 & \beta^2 & \beta^3 \end{pmatrix} \begin{pmatrix} y^1 \\ y^2 \\ y^3 \end{pmatrix} = \begin{pmatrix} x^1 & x^2 & x^3 \end{pmatrix} \begin{pmatrix} \alpha^1\beta^1 & \dots & \alpha^1\beta^3 \\ \vdots & & \vdots \\ \alpha^3\beta^1 & \dots & \alpha^3\beta^3 \end{pmatrix} \begin{pmatrix} y^1 \\ y^2 \\ y^3 \end{pmatrix}\tag{11}$$

In otherword, we are just invoking associative law(which holds in vector space):

$$(x\alpha^T)(\beta y^T) = x(\alpha^T\beta)y^T\tag{12}$$

where the outer product $\alpha^T\beta = A$.

We denote bilinear forms $x^T Ay$ in sandwich notations: $\langle x|A|y\rangle$

(Step 3) The only possble opeartor to be sandwiched without changing the result:

$$\langle x|Ay\rangle = \langle x|y\rangle\tag{13}$$

is when $A = I$. We therefore conclude that the outer product $|e_\nu\rangle\langle e_\nu|$ is the identity. In math notation, this is saying:

- For orthonormal basis $\{e_\nu\}$, we have

$$\sum_{\nu=1}^n e_\nu e_\nu^T = I\tag{14}$$

Now we consider representing $|x\rangle$ in β :

$$\begin{aligned}|x\rangle &= (|e_\nu\rangle\langle e_\nu|)|x\rangle \\ &= |e_\nu\rangle\langle e_\nu|x\rangle\end{aligned}\tag{15}$$

Now consider taking the inner product with another basis e_μ :

$$\langle e_\mu | x \rangle = \langle e_\mu | e_\nu \rangle \langle e_\nu | x \rangle \quad (16)$$

0.1.2 Infinite-dimensional Vector Space

Function Spaces

L^p Space

Definition 0.1.1 (space). L^p

Let (X, Σ, μ) be a measure space and $1 \leq p \leq \infty$. The space $L^p(X, \mu)$ is the set of all measurable functions from X to $\mathbb{R}$ or $\mathbb{C}$ such that $\int_X |f(x)|^p d\mu < \infty$.

- Note L^p is a normed space with norm:

$$\|f\|_{L^p} = \left(\int_X |f|^p d\mu \right)^{1/p} \quad (17)$$

p can be the symbol ∞ with the norm defined as:

$$\|f\|_{L^\infty} = \inf\{C \in \mathbb{R}_{\geq 0} : |f(x)| < C, \forall x\} \quad (18)$$

- It is complete, hence it is a Banach Space

Linear Operators

Definition 0.1.2 (Linear Operators). Let T be a map between two normed space X and Y . T is linear if:

$$T(x + \lambda y) = T(x) + \lambda T(y) \quad (19)$$

$$\forall x, y \in X, \lambda \in K$$

If we consider linear functions $f : \mathbb{R} \rightarrow \mathbb{R}$, it seems continuity is free since we can always write $T(x) = kx$ hence:

$$\|T(x) - T(x_0)\| = \|k(x - x_0)\| = k\|x - x_0\| \quad (20)$$

so we can easily find a delta ball of radius $\frac{\epsilon}{k}$ given an epsilon ball such that the delta ball is mapped into the epsilon ball. More generally we can define bounded operator to make the above logic work:

Definition 0.1.3 (Bounded Operators). Let T be a map between two normed space X and Y . T is bounded if $\exists C > 0$:

$$\|T(x)\|_Y \leq C\|x\|_X \quad (21)$$

$$\forall x \in X$$

Proposition 0.1.2 (Continuous and Bounded are equivalent for Linear Operator). The following are equivalent:

1. If X and Y are normed spaces and $T : X \rightarrow Y$ is a linear map which is continuous from X to Y
2. T is bounded

Proof. • Suppose T is **linear and bounded**, $\forall x_0 \in X$, $\forall \epsilon > 0$, consider the open ball around x_0 with radius ϵB_ϵ^o . Then $\forall x \in B_\epsilon^o$:

$$\begin{aligned} \|T(x) - T(x_0)\| &= \|T(x - x_0)\| \\ &\leq C\|x - x_0\| \end{aligned} \quad (22)$$

Now to make $C\|x - x_0\| < \epsilon$, we need $\|x - x_0\| < \frac{\epsilon}{C}$. Hence $\delta(\epsilon) = \frac{\epsilon}{C}$

- Suppose T is **linear and continuous**. Then in particular T is continuous at $x = 0$. We can find a $\delta > 0$ such that $\|x\| < \delta \Rightarrow \|T(x)\| < 1$. Now to control arbitrary $x \in X$, we can rewrite x as

$$x = \left(\frac{2}{\delta}\right) \|x\| \left(\frac{\delta x}{2\|x\|}\right) = C\|x\|x' \quad (23)$$

Note $\|x'\| = \frac{\delta}{2\|x\|} \|x\| < \delta$ hence $\|T(x')\| \leq 1$. By linearity $T(x) = T(C\|x\|x') = C\|x\|T(x')$. Thus:

$$\|T(x)\|_Y = C\|x\|\|T(x')\| \leq C\|x\| \quad (24)$$

□

It turns out all finite-dimensional linear operator are bounded. However, infinite-dimensional linear operator can easily be unbounded.

Example 0.1.1 (Unbounded Operator). Consider the **differential operator** $\frac{d}{dx} : C^\infty[0, 2\pi] \rightarrow C^\infty[0, 2\pi]$ Consider the family $f_n = \sin(nx)$. Now

$$\begin{aligned}\|f_n\|_{L_2}^2 &= \int_0^{2\pi} \sin^2(nx) dx \\ &= \int_0^{2\pi} \frac{1 - \cos(2x)}{2} dx \\ &= \pi\end{aligned}\tag{25}$$

However $\frac{df_n}{dx} = n \cos(nx)$ and:

$$\begin{aligned}\|f'_n\|_{L_2}^2 &= \int_0^{2\pi} n^2 \cos^2(nx) dx \\ &= n^2 \int_0^{2\pi} \frac{1 + \cos(2x)}{2} dx \\ &= n^2 \pi\end{aligned}\tag{26}$$

Hence the operator $\frac{d}{dx}$ is not bounded since suppose there is such C :

$$\|f'\|^2 \leq C\|f\|^2\tag{27}$$

We can always pick n , $n^2 > C$ such that $n^2 \pi = \|f'_n\|^2 > C\|f_n\|^2 = C\pi$.

The Uniform Boundedness Theorem

Baire Category Theorem

Theorem 0.1.3 (Baire Category Theorem). Let X be a **complete metric space** then:

1. Open dense subsets of X is closed under countable union
2. X is not a countable union of nowhere dense set

Linear Differential Operator

The Formal Linear Differential Operator We define the differential operator as:

Definition 0.1.4 (Differentiation Operator). Let $D : L^2[a, b] \rightarrow L^2[a, b]$ be the mapping defined by:

$$D(u) = \frac{d}{dx}u \quad (28)$$

for all $u \in L^2[a, b]$

Now we can define the *formal linear operator* by some operator algebra:

Definition 0.1.5 (Formal linear differential operator). Let $p(x)$ be any polynomial over $\mathbb{C}$ of positive degree:

$$p(x) = a_\nu x^\nu \quad (29)$$

where a_ν are functions of x . Define $L := p(D)$ as the **formal linear differential operator** that correspond to the **auxillary polynomial** p :

$$L = a_\nu D^\nu \quad (30)$$

The **order** of L is given by the order of p .

We call it *formal* since we want to contrast with *concrete* linear differential operator which is a formal operator endorsed with boundary conditions (This convention is used in *Mathematical Physics* by Michael Stone and Paul Goldbart). We can think of the difference between them as:

- The **nullspace of formal operator** is the space of general solution of homogenous differential equation
- The **nullspace of concrete operator** is the space of particular solution to homogenous differential equation with boundary conditions

We'll explore the nullspace of formal operators with a bit more detail now.

Nullspace of formal operator We first observe something very nice about solutions to homogenous differential equations: they belong to C^∞ (infinitely differentiable).

The Bootstrapping Lemma

Lemma 0.1.4. Let L be a formal operator of order n . Suppose the coefficients of the auxillary polynomial are k times continuously differentiable ($a_\nu \in C^k$) then $\forall y \in \text{Null}(L)$, y is at least $k + n$ times continuously differentiable.

Proof. Suppose $y \in \text{Null}(L)$, note trivially $y \in C^n$. Then:

$$a_0 y + a_1 y^{(1)} + a_2 y^{(2)} \dots + a_n y^{(n)} = 0$$

$$y^{(n)} = -(b_0 y + b_1 y^{(1)} + b_2 y^{(2)} \dots \underbrace{b_{n-1} y^{(n-1)}}_{\in C^{\min(k,1)}}) \quad (31)$$

Where $b_\nu = a_\nu / a_n$. Now $b_\nu \in C^k$ by closedness under polynomial division. Also $y^{(\nu)} \in C^{n-\nu}$ hence the R.H.S is at least one time continuously differentiable (the bottle neck being $a_{n-1} y^{(n-1)} \in C^{\min(k,1)}$). Therefore $y^{(n)} \in C^1$. But this implies $y \in C^{n+1}$ hence we repeat the above argument to get $y^{(\nu)} \in C^{n+1-\nu}$. The R.H.S has the bottleneck $a_{n-1} y^{(n-1)} \in C^{\min(k, n+1-(n-1))} = C^{\min(k,2)}$. We can repeat this procedure m times and yield that $y^{(n)} \in C^{\min(k,m)}$. This means the best we can get is $y^{(n)} \in C^k$ or $y \in C^{m+k}$ (after which the coefficient becomes the bottleneck). □

Corollary 0.1.4.1. If the coefficients are in C^∞ then $y \in \text{Null}(L)$ are in C^∞ .

Nullspace of L is an n -dimensional subspace of C^∞

Linear Independence: Wronskian

Solving $Ly = 0$ with constant coefficients

Exercise 1 (1-dimensional Scattering Problem)

Consider the 1d Schrodinger equation:

$$-\frac{d^2\psi}{dx^2} + V\psi = E\psi \quad (32)$$

This is an eigenvalue problem of the operator $H = -\frac{d^2}{dx^2} + V$:

$$H\psi = E\psi \quad (33)$$

Where V has finite support $[-a, a]$, $a \in \mathbb{R}$. Denote:

1. $L := \{x \in \mathbb{R} : x < -a\}$
2. $R := \{x \in \mathbb{R} : x > a\}$

Problems:

1. Show that in L, R , the general solution to the formal operator $H - E$ where $E = k^2$ is:

$$\begin{aligned} \psi_k &= A \exp(ikx) + B \exp(-ikx) & x \in L \\ \psi_k &= C \exp(ikx) + D \exp(-ikx) & x \in R \end{aligned} \quad (34)$$

2. Now consider two asymptotic boundary condition:

$$\begin{cases} \Pi_L : D = 0, x \rightarrow \infty \\ \Pi_R : C = 0, x \rightarrow -\infty \end{cases} \quad (35)$$

where Π_L depicts a left-incident wave with no incoming wave from the right and Π_R is the opposite. Show that the solution(in L, R) for the BC Π_1 and $k > 0$ is:

$$\psi_k^L(x) = \begin{cases} \exp(ikx) + r_L(k) \exp(-ikx) & x \in L \\ t_L(k) \exp(ikx) & x \in R \end{cases} \quad (36)$$

Similarly, for Π_2 , $k < 0$:

$$\psi_k^R(x) = \begin{cases} t_R(k) \exp(ikx) & x \in L \\ \exp(ikx) + r_R(k) \exp(-ikx) & x \in R \end{cases} \quad (37)$$

where t, r denotes the *transmission* and *reflection* coefficient.

3. Note that ψ_k^* is also a solution. Use properties of Wronskian to show that:

$$|t_{L/R}|^2 + |r_{L/R}|^2 = 1 \quad (38)$$

Solution to Exercise 1

1. Fix a k . Consider the basis ψ_k^L, ψ_k^{L*} . Form the Wronskian in L :

$$W_L = \begin{vmatrix} \exp(ikx) + r_L \exp(-ikx) & \exp(-ikx) + r_L^* \exp(ikx) \\ ik \exp(ikx) - ikr_L \exp(-ikx) & -ik \exp(-ikx) + ikr_L^* \exp(ikx) \end{vmatrix} \quad (39)$$

Evaluate the determinant:

$$\begin{aligned} W_L &= (-ik) - ikr_L \exp(-2ikx) + ikr_L \exp(2ikx) + ik|r_L|^2 \\ &\quad - [ik - ikr_L \exp(-2ikx) + ikr_L \exp(2ikx) - ik|r_L|^2] \\ &= 2ik(|r_L|^2 - 1) \end{aligned} \quad (40)$$

Next form Wronskian in R :

$$W_R = \begin{vmatrix} t_L \exp(ikx) & t_L \exp(-ikx) \\ ikt_L \exp(ikx) & -ikt_L \exp(-ikx) \end{vmatrix} = -2ik|t_L|^2 \quad (41)$$

Since we would expect $W_R = W_L = 0$, hence:

$$\begin{aligned} W_R - W_L &= 0 \\ 2ik(|r_L|^2 + |t_L|^2 - 1) &= 0 \end{aligned} \quad (42)$$

Since $k > 0$ we get as desired.

4. Fix k . We repeat the same procedure with $\{\psi_k^L, \psi_{(-k)}^R\}$. We evaluate W_L :

$$W_L = \begin{vmatrix} \exp(ikx) + r_L(k) \exp(-ikx) & t_R(-k) \exp(-ikx) \\ ik \exp(ikx) - ikr_L(k) \exp(-ikx) & -ikt_R(-k) \exp(-ikx) \end{vmatrix} = -2ikt_R(-k) \quad (43)$$

Similarly,

$$W_R = -2ikt_L(k) \quad (44)$$

Again, $W_L = W_R = 0$ so we get as desired.

$Ly = 0$ **with general** $a_\nu(x)$

Inner Product Space

Distribution and Test Functions

Continuous Indexing

Duality Pairing

Exercise 2 (δ is a ring homomorphism)

Consider $\mathcal{T}$ and $\mathbb{R}$ as two rings.

1. Prove that δ preserves ring multiplication

$$f\delta(x) = f(0)\delta(x) \quad (45)$$

2. Distribution Derivative of $f\delta$

Show that the distribution derivative of $f\delta$ is

$$\delta[-fg'] = f(0)\delta'[g] \quad (46)$$

or, alternatively,

$$\delta[-fg'] = \delta[f'g] + \delta'[fg] \quad (47)$$

Solution to Exercise 2

1. $\forall g \in \mathcal{T}$, we have:

$$\begin{aligned} (f\delta)[g] &= \delta[fg] \\ &= f(0)g(0) \\ &= f(0)\delta \end{aligned} \quad (48)$$

As desired.

2. We can directly compute the derivative:

$$\delta[f\phi] = f(0)\delta[\phi] \xrightarrow{\phi=-g'} \delta[f(-g')] = f(0)\delta[-g'] = f(0)\delta'(g) \quad (49)$$

where the last equality used the definition of distribution derivative. Now we show that $\delta[f'g] + \delta'[fg] = f(0)\delta'(g)$. We use ring homomorphism property to establish:

$$\begin{aligned} \delta[f'g] + \delta'[fg] &= \delta[f']\delta[g] + \delta[-(fg')] \\ &= \delta[f']\delta[g] - (\delta[f']\delta[g] + \delta[f]\delta[g']) \\ &= -\underbrace{\delta[f]}_{f(0)}\delta[g'] \\ &= f(0)\delta'[g] \end{aligned} \quad (50)$$

Exercise 3 ($\delta^{(1/2)}$)

We can define the distribution $\delta^{(1/2)}$ as a fourier transform of $|k|^{1/2}$:

$$\delta^{(1/2)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |k|^{1/2} \exp(ikx) dk \quad (51)$$

The fourier transform is clearly divergent, hence we can add a regulator. Define:

$$\begin{aligned} \delta_{\mu}^{(1/2)} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |k|^{1/2} \exp(ikx) \exp(-\mu|k|) dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |k|^{1/2} \exp(ikx - \mu|k|) dk \end{aligned} \quad (52)$$

1. Evaluate the integral get a closed form expression of $\delta_{\mu}^{(1/2)}$ by:

- Splitting $k < 0$ and $k > 0$
- Change variable to $y = |k|^{1/2}$
- Write the result as $\frac{d}{d\mu}$ of a Gaussian integral
- Use the trigonometric relation

$$\arctan(z) = \frac{i}{2} \log \left(\frac{1 - iz}{1 + iz} \right) \quad (53)$$

2. Investigate $\delta_{\mu}^{(1/2)}$ by:

- Sketch $\delta_{\mu}^{(1/2)}$ for different μ
- Compute the zeros of $\delta_{\mu}^{(1/2)}$

3. Show that $\delta_{\mu}^{(1/2)}$ is the fourier transform of a function that vanishes at $k = 0$. What property of $\delta_{\mu}^{(1/2)}$ can we infer?

4. Compute the limit $\lim_{\mu \rightarrow 0^+} \delta_{\mu}^{(1/2)}$

5. Assuming that for $|x| > \max |\{x_0 : \delta_{\mu}^{(1/2)}(x_0) = 0\}|$, the convergence of

$$\delta_{\mu}^{(1/2)} \rightarrow \delta^{(1/2)} \quad (54)$$

is pointwise. Deduce:

$$\int_{-\infty}^{\infty} \delta^{(1/2)} = -\sqrt{\frac{1}{8\pi}} \int_{-\infty}^{\infty} \frac{1}{|x|^{3/2}} \{\phi(x) - \phi(0)\} dx \quad (55)$$

Solution to Exercise 3

1. Following the guide:

$$\begin{aligned} \delta_{\mu}^{(1/2)} &= \frac{1}{2\pi} \left\{ \int_{-\infty}^0 (-k)^{1/2} \exp(ikx + \mu k) dk + \int_0^{\infty} (k)^{1/2} \exp(ikx - \mu k) dk \right\} \\ &= \frac{1}{2\pi} \left\{ \underbrace{\int_{-\infty}^0 y \exp(-iy^2 x - \mu y^2) dy}_A + \underbrace{\int_0^{\infty} y \exp(iy^2 x - \mu y^2) dy}_B \right\} \end{aligned} \quad (56)$$

Exercises

Formal adjoint

Concrete Adjoint

The Sturm-Liouville Operator The **Sturm-Liouville operator** is a linear differential operator of the form

$$\mathcal{L} = -\frac{d}{dx} \left(p \frac{d}{dx} y \right) + q \frac{d}{dx} \quad (57)$$

We've shown that the Sturm-Liouville operator is self-adjoint with appropriate boundary condition in the sense that:

$$\langle u, \mathcal{L}v \rangle = \langle \mathcal{L}^*u, v \rangle \quad (58)$$

hence it is analogous to self-adjoint operator in finite-dimensional vector spaces, which as symmetric matrix representation when the underlying field is real. We now make this connection explicit, i.e., actually constructing the discretized, finite-dimensional matrix representation of the Sturm-Liouville Operator.

A Discrete Analog Let's consider a concrete SL operator defined on $[0, 1]$ with Dirichlet boundary condition $y(0) = y(1) = 0$. Instead of directly discretizing $\mathcal{L}$, we aim to find a matrix L such that discretizes the quadratic form:

$$F(y) = \int y \mathcal{L} y dx \rightarrow \hat{y}^T L \hat{y} \quad (59)$$

First we note that:

$$\begin{aligned} F(y) &= \int_0^1 y \left(-\frac{d}{dx} \left(p \frac{d}{dx} y \right) + qy \right) dx \\ &= \int_0^1 -y \frac{d}{dx} \left(p \frac{d}{dx} y \right) dx + \int_0^1 qy^2 dx \\ &= - \left\{ \left[yp \frac{dy}{dx} \right]_0^1 - \int p \left(\frac{dy}{dx} \right)^2 \right\} + \int_0^1 qy^2 dx \\ &= \int_0^1 py'^2 + qy^2 \end{aligned} \quad (60)$$

Now we discretize y, p, q to N -dimensional arrays:

$$\begin{aligned} \hat{y} &= [y_1, \dots, y_N] \\ \hat{p} &= [p_1, \dots, p_N] \\ \hat{q} &= [q_1, \dots, q_N] \end{aligned} \quad (61)$$

Using forward difference we get:

$$\hat{y} = \left(\frac{y_{i+1} - y_i}{h} \right)^2 \quad (62)$$

Rewrite the integral as a finite sum we obtain a discretized version of (60):

$$\hat{F}(\hat{y}) = \sum_{i=1}^N h p_i \left(\frac{y_{i+1} - y_i}{h} \right)^2 + q_i y_i^2 \quad (63)$$

Our goal is now to find a matrix L such that $\hat{y}^T L \hat{y} = \hat{F}(\hat{y})$. Rewrite $\hat{F}(\hat{y})$:

$$\hat{F}(\hat{y}) = \sum_{i=1}^N N p_i (y_{i+1}^2 - 2y_{i+1}y_i + y_i^2) + \frac{1}{N} q_i y_i^2 \quad (64)$$

We can split $L = L_1 + L_2$ such that

1. $\hat{y}^T L_1 \hat{y} = \sum_{i=1}^N N p_i (y_{i+1}^2 - 2y_{i+1}y_i + y_i^2)$
2. $\hat{y}^T L_2 \hat{y} = \sum_{i=1}^N \frac{1}{N} q_i y_i^2$.

(Constructing L_1) Let's first deal with L_1 . Consider the 2×2 block with the upperleft corner at (i, i) that should represent $N p_i (y_{i+1}^2 - 2y_{i+1}y_i + y_i^2)$. This is easy:

$$A_i = N p_i \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (65)$$

To assemble L_1 we can overlap A_i along the diagonal of L with a stride of 1. Concretely:

```
for i=1,...,N -1
    L[i,i] += Np_i
    L[i+1,i+1] += Np_i
    L[i,i+1] -= Np_i
    L[i+1,i] -= Np_i
```

Example 0.1.2. A quick example for $N = 4$:

$$4 \begin{pmatrix} p_1 & -p_1 & 0 & 0 \\ -p_1 & p_1 + p_2 & -p_2 & 0 \\ 0 & -p_2 & p_2 + p_3 & -p_3 \\ 0 & 0 & -p_3 & p_3 \end{pmatrix} \quad (66)$$

(Constructing L_2) L_2 is purely diagonal and is easier to derive. Since it only has quadratic term y_i^2 , at each row i we only need to include y_i . Hence:

$$L_2 = \frac{1}{N} \begin{pmatrix} q_1 & & \\ & \ddots & \\ & & q_N \end{pmatrix} \quad (67)$$

(Assembly $L_1 + L_2$) It is clear that L should look like:

$$L = N \begin{pmatrix} p_1 & -p_1 & & & \\ -p_1 & p_1 + p_2 & -p_2 & & \\ & \ddots & \ddots & \ddots & \\ & & -p_{N-1} & p_{N-1} + p_N & -p_N \\ & & & -p_N & p_N \end{pmatrix} + \frac{1}{N} \begin{pmatrix} q_1 & & & & \\ & \ddots & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & q_N \end{pmatrix} \quad (68)$$

However, since we are dealing with concrete operators with accompanying domain and boundary condition, we need to encooperate the Dirchelt boundary condition $y(0) = 1, y(1) = 0$ into L . To do so we need to zero-out the first row and column of L . Hence finally we get:

Proposition 0.1.3. The Sturm-Liouville Operator on $[0, 1]$ with Dirchlet boundary condition $y(0) = y(1)$ can be discretized as

$$L = N \begin{pmatrix} 0 & 0 & & & \\ 0 & p_1 + p_2 & -p_2 & & \\ & -p_2 & \ddots & \ddots & -p_{N-1} \\ & & \ddots & -p_{N-1} & p_{N-1} + p_N & 0 \\ & & & 0 & 0 \end{pmatrix} + \frac{1}{N} \begin{pmatrix} 0 & & & & \\ & q_2 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & q_{N-1} & 0 \end{pmatrix} \quad (69)$$

We can see that L is indeed a symmetric real matrix(Hermitian)!

Numpy Implementation This can easily be translated to naive numpy code:

```

class SturmLiouville:
    def __init__(self, N, p, q=None, grid='forward'):
        A = np.zeros(shape=(N,N))
        if grid == 'forward':
            for i in range(N -1):
                A[i, i] += p[i]
                A[i+1, i+1] += p[i]
                A[i, i+1] -= p[i]
                A[i+1, i] -= p[i]

        if q is not None:
            I = np.eye(N)
            L = A + q @ I
        else:
            L = A

        # Dirchlet Boundary Condition
        L[0,0]=0.0
        L[0,1]=0.0
        L[1,0]=0.0
        L[N -1,N -1]=0.0
        L[N -1,N -2]=0.0
        L[N -2,N -1]=0.0
        self.L = (L * N ** 2)
        self.N = N

    def solve_eigen(self, top_k=10):
        eval, evec = np.linalg.eig(self.L)
        top_k_idx = np.argsort(eval[eval>0])[:top_k]
        return eval[top_k_idx] , evec[:, top_k_idx], top_k_idx

```

Example: $-\frac{d^2}{dx^2}$ as a Matrix We can do a quick example to showcase that L is indeed a discretization of $\mathcal{L}$.

Example 0.1.3 (-). $\frac{d^2}{dx^2}$

We have already seen that the operator:

$$T = -\frac{d^2}{dx^2}, D(T) = \{y, Ty \in L^2[0, 1] : y(0) = y(1) = 0\} \quad (70)$$

has eigenvalues and eigenvectors:

$$\begin{aligned} y_j(x) &= \sqrt{2} \sin j\pi x \\ \lambda_j &= j^2\pi^2 \end{aligned} \quad (71)$$

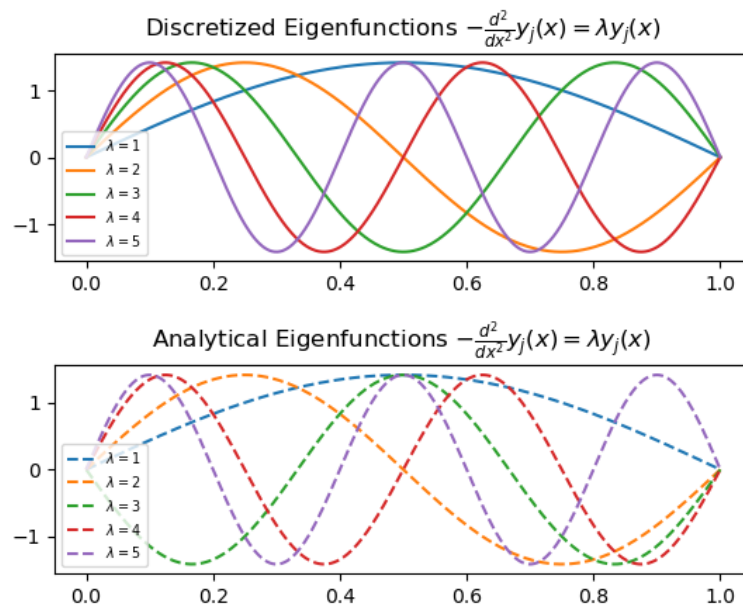
for $j = 1, 2, 3, \dots$. The problem is that, can we solve a matrix eigenvalue problem to recover exactly the same result? Indeed we can! Passing $p = -1$ and $q = 0$ gets us:

$$L = N \begin{pmatrix} 0 & 0 & & & \\ 0 & 2 & -1 & & \\ & -1 & \ddots & \ddots & \\ & & \ddots & 2 & -1 \\ & & & -1 & 2 & 0 \\ & & & & 0 & 0 \end{pmatrix} \quad (72)$$

and solving the matrix eigenvalue problem

$$L\hat{y} = \lambda\hat{y} \quad (73)$$

gets us: ¹



Exercise 4

1. We discretized L based on the quadratic form $\int_0^1 y \mathcal{L} y dx$, hence $\hat{y}^T L \hat{y}$ should coincide with $\int_0^1 y \mathcal{L} y dx$. Come up with at least 2 functions and 2 operators that satisfy the requirement ($L^2(0,1), y(0) = y(1) = 0$). Compute the quadratic form analytically and numerically. Check that the results are the same.
2. Remove the masking code snippet (which implements Dirichlet boundary condition) in the naive SL class `SturmLiouville`. What is the resulting eigenvectors? Explain the result.
3. Try coming up with different discretization with center difference or back difference. Implement it in `python` or `julia`.
 - Is the resulting matrix L still symmetric?
 - Center differencing scheme results in a numerical artifact, what is the numerical artifact? Try explaining why such artifact occur (*Hint: Note that even and odd indices are decoupled*)

¹Since iterative eigensolvers gives eigenvectors of arbitrary magnitude $v_j = a \sin(j\pi(x))$, we need to normalize properly by 1. Ensure $a > 0$

1. Divide by $\sqrt{\sum y_i^2 h}$

0.2 Calculus of Variation

0.2.1 Euler-Lagrange Equation

Differential Equation from Physics

A lot of physical science and engineering is about finding a reduction from a physical problem to differential equations. The newtownian way is to zoom into some local point of the system and derive a set of constraint that holds in the neighborhood. In the case of the vibration of a string, for example, we would find some local segment of string and analyze the tension that acts on it:

$$T(\sin(\theta_1) - \sin(\theta_2)) = m \frac{\partial^2 u}{\partial t^2} \quad (74)$$

where u is the vertical displacement from equilibrium. We then take the calculus limit and find out that locally the acceleeration of the vertical displacement should be proportional to the curvature. Hence the one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} \quad (75)$$

2

The Lagrangian way is the opposite, we recognize some global objective the system need to be optimized toward, i.e., minimizing the lagragian $L = T - V$ where T is *kinetic energy* and V is *potential energy*. Then, by introduce some perturbation, we extract a constraint which also manifest itself as a differential equation. The following sections descirbes this process in detail.

Newtonian and Lagrangian Mechanics

- Newtonian mechanics compute the balance of forces locally and obtain differential equation through

$$F = m \frac{d^2 \mathbf{x}}{dt^2} \quad (76)$$

- Lagrangian mechanics translate the physics into some global optimization problem and reduce to differential equation through the Euler-Lagrange equation

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} = 0 \quad (77)$$

Functional : A global optimization objective

A functional in *calculus of variation* is a map from a function to a real number, i.e., $J : C^\infty \rightarrow \mathbb{R}$. In particular, it mostly comes in some sort of a integral. The simplest form being analgous to that of the lagrangian:

$$J[y] = \int_{x_1}^{x_2} f(x, y, y') dx \quad (78)$$

The above example suffice to establish the explict connection of (78)to Lagrangian:

Example 0.2.1 (The lagrangian for the vibrating string). *Let u be deviation from the equilibrium position*

1. The kinetic energy:

²Here a is a physical parameter intrinsic to the string, i.e., $\sqrt{T/\rho}$ where T is the tension at equilibrium adn ρ is the density.

$$T = \frac{1}{2} \int_0^L \rho u_t(x, t)^2 dx \quad (79)$$

where L is the length of the string

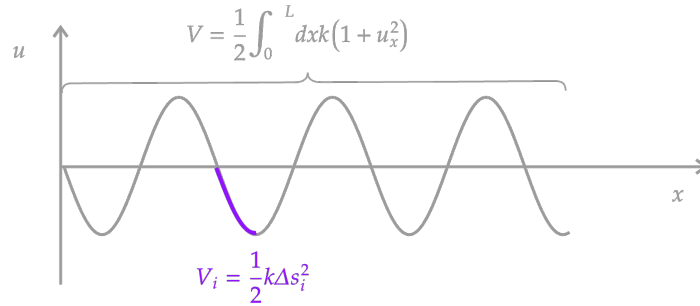
2. The potential energy:

$$V = \frac{1}{2} k \int_0^L (1 + u_x^2) dx \quad (80)$$

3

The lagrangian is hence:

$$L = \int_0^L \frac{1}{2} \left(\rho u_t(x, t)^2 - k(1 + u_x^2) \right) dx \quad (83)$$



The potential energy of a vibrating string.

We now scrutinize how exactly we optimize (78).

Euler-Lagrange Equation

Theorem 0.2.1 (Euler-Lagrange Equation). *y is a stationary point for*

$$J[y] = \int_{x_1}^{x_2} f(x, y, y^{(1)}, \dots, y^{(n)}) dx \quad (84)$$

iff:

$$\boxed{\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0} \quad (85)$$

The boxed equation is known as the **Euler-Lagrange Equation**

³It is tempting for some of us who are not trained as a physics student to write the potential energy as $\frac{1}{2} \left(\int_a^b \sqrt{1 + u_x^2} dx \right)^2$. The rationale is as follows: if we pause at a moment and straighten the vibrating string as a line. Then we observe how much elongation had happened and use hooke's law on the entire string. This is physically wrong since the elongation happens inhomogenously hence we have to apply hooke's law locally and integrate. Also mathematically,

$$\left(\int f \right)^2 = \int f^2 \quad (81)$$

iff f is some constant function. The general relationship can be obtained from Cauchy-Schwarz inequality by taking $f = \sqrt{1 + u_x^2}$ and $g = 1$

$$\left(\int_0^L \sqrt{1 + u_x^2} dx \right)^2 \leq \int_0^L 1 + u_x^2 dx \quad (82)$$

Discretized Euler-Lagrange Equation

First Integral

Definition 0.2.1 (First Integral). *Suppose*

$$y^{(n)} = f(x, y, y^{(1)}, \dots, y^{(n-1)}) \quad (86)$$

is an ordinary differential equation of order n . We can define an equivalent n -dimensional system by letting:

$$\begin{aligned} y_1 &= y \\ y_2 &= y^{(1)} \\ &\vdots \\ y_n &= y^{(n-1)} \end{aligned} \quad (87)$$

Denote by:

$$\frac{dy_i}{dx} = f_i(x, y_1, \dots, y_n) \quad (88)$$

Now suppose f is:

1. *Continuous about $x, y_1, \dots, y_n$*
2. *Continuously differentialbe about $y_1, \dots, y_n$*

Define the first integral of the n^{th} -order problem to be a function $I = I(x, y_1, \dots, y_n)$ that also satisfies the above two requirements, and furthermore has the definitive property:

$$I(x, y_1, \dots, y_n) = \text{Constant} \quad (89)$$

along solution curves $\Gamma = (y_1(x), y_2(x), \dots, y_n(x))$ with the constant determined by the solution curve.

We are confronted by the higher-order ordinary differential equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (90)$$

The compute the first integral of this equation is to rewrite the above to obtain an equation of the form:

$$\frac{d}{dt} I(x, y, y', \dots) = 0 \quad (91)$$

The strategy is to compute the difference between (90) and the exact differential:

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \sum_{j \geq 0} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial y^{(1)}} \frac{dy^{(1)}}{dx} + \dots \quad (92)$$

If we multiply (90) by y' and subtract the two equations we get:

$$\frac{df}{dx} - y' \left\{ \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right\} = \frac{\partial f}{\partial x} + y'' \frac{\partial f}{\partial y'} + y' \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) - \sum_{j \geq 2} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} \quad (93)$$

The particular form that allows easy first integral

If we assume f is only dependent on y, y' , the above becomes $f(y, y')$, then

$$\begin{aligned} \sum_{j \geq 2} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} &= 0 \\ \frac{\partial f}{\partial x} &= 0 \end{aligned} \quad (94)$$